

LIN 313 Language and Computers

Word Vectors

Date: 10/31/2025

Name: _____

Vector Analogies

Instructions: In this lab we will be editing code. Navigate to the Google Colab notebook linked from today's lecture on the website. You have to be logged in to a Google account.

1. Find the 5 nearest neighbors of the word **pineapple**
 - a. Expand the neighborhood size to 10. What are the neighbors 6-10?

2. Pick a word and find the nearest neighbors. List both.
 - a. List off the neighbors (but not the word!) to someone else in the class. Can they guess what the word is? Try a few different words (you don't have to write these down)

3. Have word2vec solve the analogy $\text{space} : \text{time} :: \text{land} : \underline{\hspace{2cm}}$. (Space is to time as land is to $\underline{\hspace{2cm}}$). Hint: The vector arithmetic for this space + land - time

4. Come up with a few more analogies and see if word2vec can solve them. Write down one or two of the more interesting ones, along with the "answer" given by word2vec. Is the word you expect? Is the answer in the top five?